Bayesian Inference for Speech Recognition in Noisy Environments

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Bayesian Research Presentation

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Introduction

- Speech recognition in noisy environments is hindered by distorted acoustic features.
- Bayesian inference provides a principled statistical framework for refining phoneme feature estimates.
- **Objectives:**
  - Assess the impact of noise variance on posterior estimates.
  - Extend the model to account for phoneme-specific noise levels.
  - Compare Bayesian methodologies with baseline recognition techniques.

Bayesian Model Specification

Prior and Likelihood Specification:

$$
\begin{aligned}
\mu_j &\sim \mathcal{N}(\mu_0, \Sigma_0), \\
y|\mu_j &\sim \mathcal{N}(\mu_j, \Sigma),
\end{aligned}
$$

where μ_0 and Σ_0 are derived from high-quality recordings.

Inference Techniques

- **Direct Monte Carlo Sampling:** Samples drawn from posterior:

$$\mu_j | y \sim \mathcal{N}(\mu_{post}, \Sigma_{post})$$

where:

$$\Sigma_{post} = (\Sigma_0^{-1} + \Sigma^{-1})^{-1},$$
$$\mu_{post} = \Sigma_{post}(\Sigma_0^{-1}\mu_0 + \Sigma^{-1} y)$$

- **Gibbs Sampling:** Iterative conditional sampling.

Model Extension: Phoneme-Specific Noise

- Noise covariance Σ_j varies across phonemes.
- Modified observation model:

$$y|\mu_j \sim \mathcal{N}(\mu_j, \Sigma_j)$$

- **Evaluation of Recognition Accuracy:**
- Bayesian approach is compared with baseline methods (Maximum Likelihood Estimation) under varying noise conditions

$$\hat{\mu}_{MLE} = \arg\max_{\mu} P(y \mid \mu)$$

$$y \sim \mathcal{N}(\mu, \Sigma)$$

Simulation Results: Effect of Noise Variance

- m_u = 2.8 to 4.6, prior mean for the feature vector
- Posterior mean estimates remain stable under noise variation (Σ from 0.1 to 1.0):
  - Monte Carlo Estimates: Mean X ranged from **2.7946** to **2.8065**, Mean Y from **4.5911** to **4.6084**.
  - Gibbs Sampling Estimates: Mean X varied between **2.7953** and **2.8061**, Mean Y between **4.5933** and **4.6045**.
- Posterior covariance increases with noise, indicating growing uncertainty.

Σ	Monte Carlo Covariance	Gibbs Covariance
0.1	$\begin{bmatrix} 1.1019 & 0.0019 \\ 0.0019 & 1.0981 \end{bmatrix}$	$\begin{bmatrix} 0.0991 & 0.0002 \\ 0.0002 & 0.1003 \end{bmatrix}$
1.0	$\begin{bmatrix} 2.0063 & -0.0022 \\ -0.0022 & 2.0066 \end{bmatrix}$	$\begin{bmatrix} 1.0023 & -0.0016 \\ -0.0016 & 0.9984 \end{bmatrix}$

Cont'd

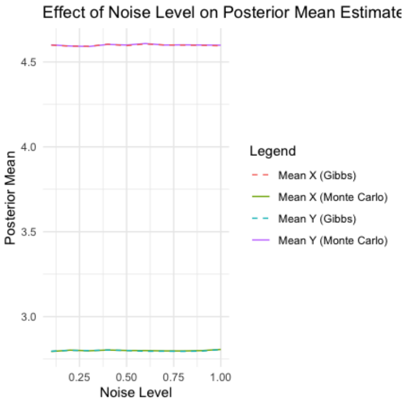

Figure: Posterior mean estimates remain stable despite increasing noise levels.

Recognition Accuracy Comparison

Σ	Bayesian RMSE	Baseline RMSE
0.1	2.1399	2.1394
0.5	2.2689	2.2681
1.0	2.4278	2.4259

- Baseline Model approach exhibits slightly superior performance in high-noise conditions.

Cont'd

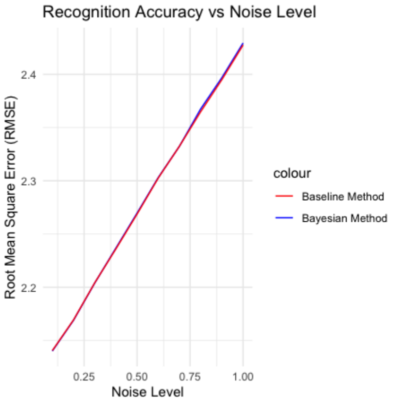

Figure: Baseline Model (Red) approach exhibits slightly superior performance in high-noise conditions than bayesian (blue).

Conclusion

- Bayesian inference provides robust phoneme feature estimates in noisy conditions.
- Posterior means remain stable; uncertainty in estimates increases with noise variance.
- Baseline methods slightly improve recognition accuracy compared to bayesian models.

References

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